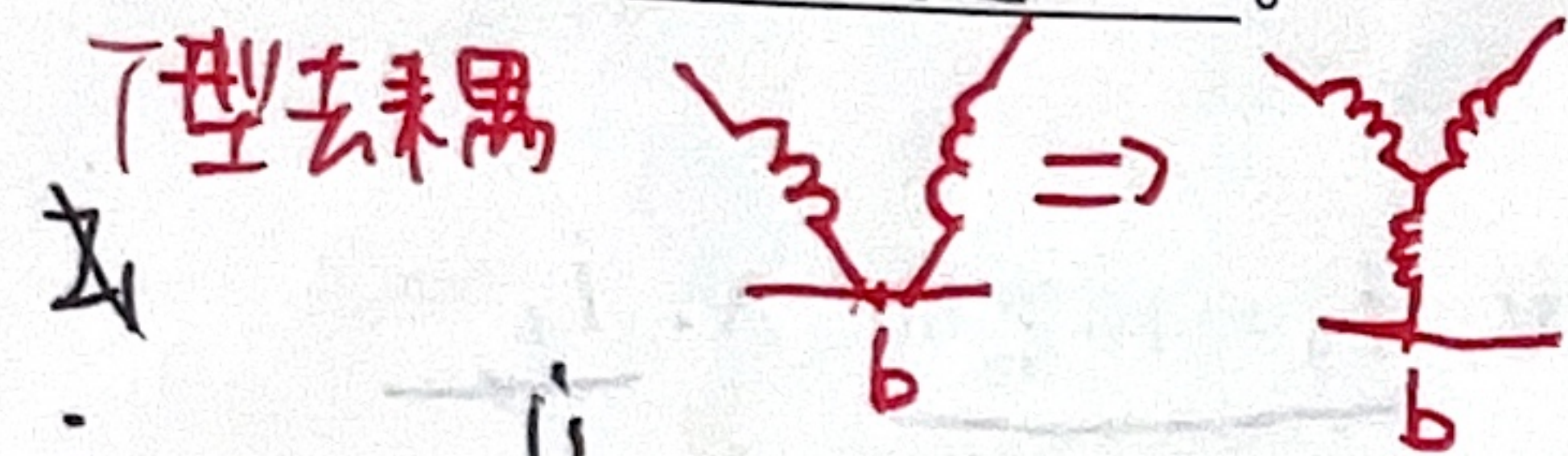
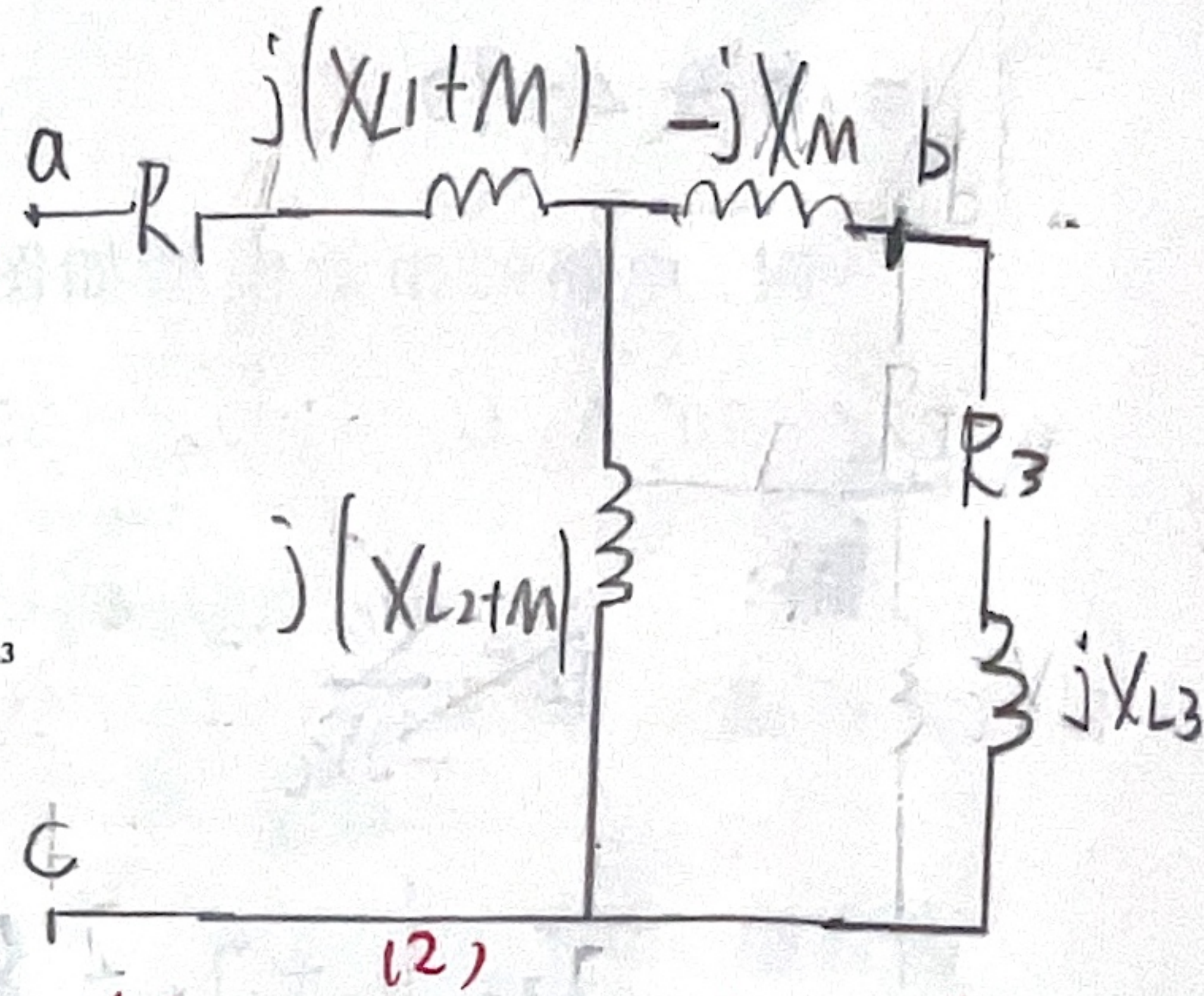
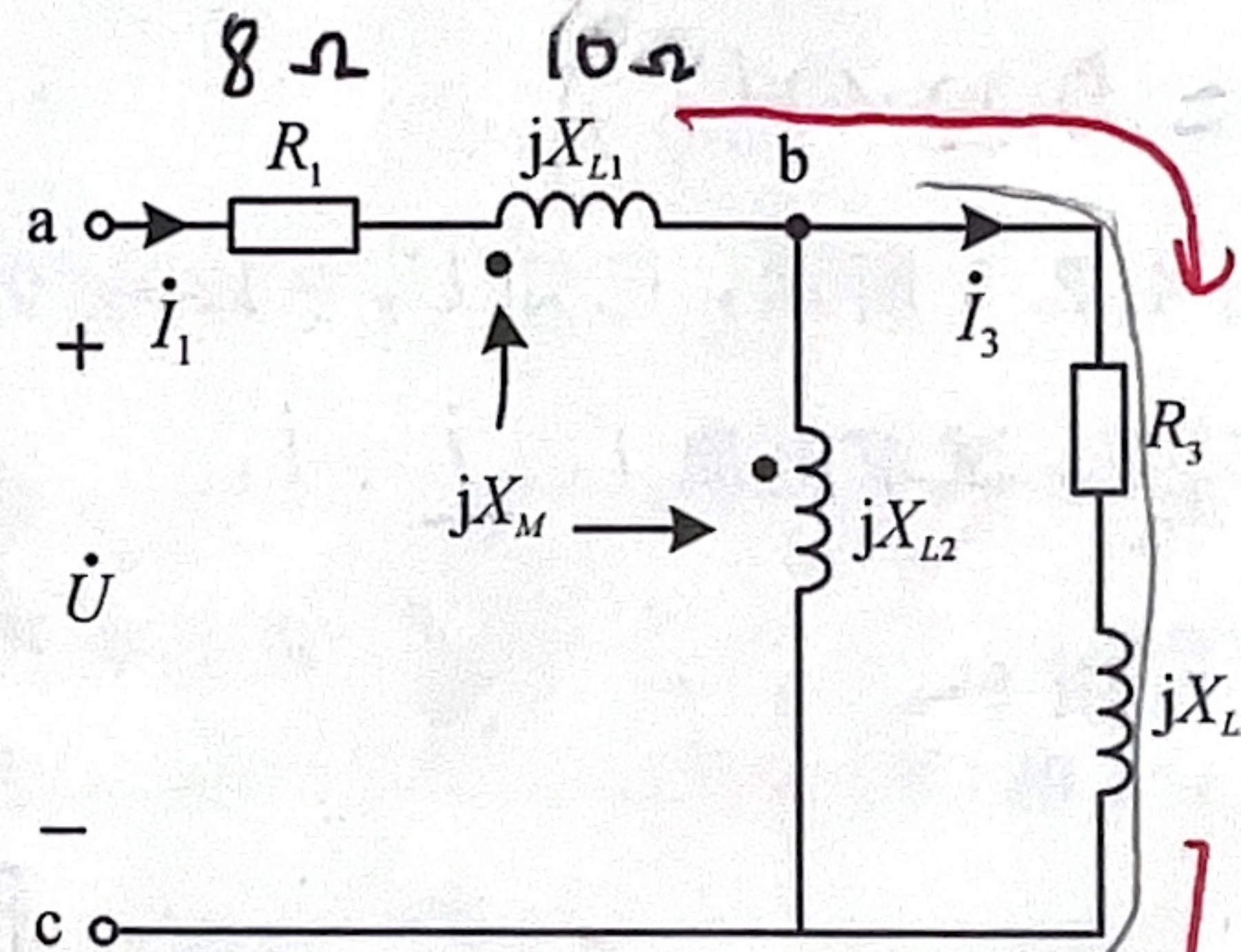


2. 电路如图 4.7.2 所示, 已知 $X_{L1} = 10 \Omega$, $X_{L2} = 6 \Omega$, $X_M = 4 \Omega$, $X_{L3} = 4 \Omega$, $R_1 = 8 \Omega$, $R_3 = 5 \Omega$, 端电压 $U = 100 \text{ V}$, 则 $\dot{I}_1 = \underline{51 - 53^\circ}$, $\dot{I}_3 = \underline{4 - 2j}$, $\dot{U}_{ab} = \underline{72 - 6j}$ 。



$\dot{I}_3 = \frac{\dot{U}}{R_1 + j(X_{L1} + M) + [j(X_{L2} + M) // -jX_M + R_3 + jX_{L3}]}$



在图(I)中, $\dot{U}_{ab} = \dot{I}_1 R_1 + L_1 \frac{d\dot{i}_1}{dt} + M \frac{d\dot{i}_3}{dt}$ 不方便算, 图 4.7.2
 ~~$\dot{U}_{ab} = \dot{I}_1 R_1 + L_1 \frac{d\dot{i}_1}{dt}$~~ 由 KVL 找条无耦合方便算的, $\dot{U} - \dot{I}_3 (R_3 + jX_{L3})$
 二、电路如图 4.7.3 所示, 已知 $R_1 = 10 \Omega$, $R_2 = 2 \Omega$, $X_{L1} = 30 \Omega$, $X_{L2} = 8 \Omega$, $X_M =$